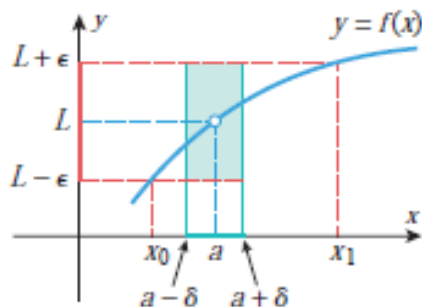


3.2 Limits and Continuity

Recall the definition of a limit from Calculus I.

Definition: Let f be a function whose domain D includes points arbitrarily close to a . Then, we say that the limit of f as x approaches a is L and we write $\lim_{x \rightarrow a} f(x) = L$ if...

for every $\varepsilon > 0$ there is a corresponding $\delta > 0$ such that if $|x - a| < \delta$ then $|f(x) - L| < \varepsilon$.



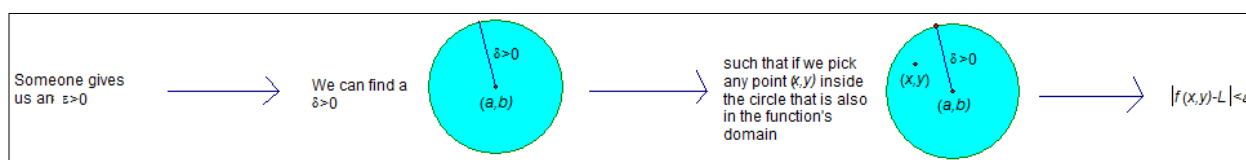
What this meant was that we needed to guarantee that a δ existed for every ε so that whenever $x \in (a - \delta, a + \delta)$ it would be true that $f(x) \in (L - \varepsilon, L + \varepsilon)$.

As we will discover, limits of functions of several variables is significantly more complicated than the limits of functions with one variable.

Definition: Let f be a function of two variables whose domain D includes points arbitrarily close to (a, b) . Then, we say that the limit of f as (x, y) approaches (a, b) is L and we write $\lim_{(x, y) \rightarrow (a, b)} f(x, y) = L$ if...

for every $\varepsilon > 0$ there is a corresponding $\delta > 0$ such that if $(x, y) \in D$ and $0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta$ then $|f(x, y) - L| < \varepsilon$.

Explanation of the Definition: This definition is saying that if $\lim_{(x, y) \rightarrow (a, b)} f(x, y) = L$, then if someone challenges us with an $\varepsilon > 0$ (think of epsilon as small) we can find a circle centered at (a, b) with radius δ such that every point in the domain that is inside the circle (not including (a, b)) produces a range value within ε of L when plugged into $f(x, y)$. An illustration of this idea is below:

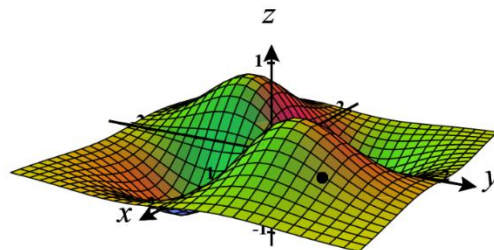


Question to the Class: In Calculus I and II when we dealt with limits like $\lim_{x \rightarrow a} f(x)$, from how many directions could we approach a ?

There are two possible directions: the left and the right. Moreover, the limits from both directions need to agree in order for the limit to exist.

Question to the Class: When it comes to limits of the form $\lim_{(x,y) \rightarrow (a,b)} f(x,y)$, from how many directions can we approach (a,b) ?

There are infinitely many directions from which we can approach (a,b) , and in order for the limit to exist, the limit needs to be the same from ALL possible directions!



Remark: If ...

$f(x,y) \rightarrow L_1$ as $(x,y) \rightarrow (a,b)$ along a path C_1 and

$f(x,y) \rightarrow L_2$ as $(x,y) \rightarrow (a,b)$ along a path C_2 where $L_1 \neq L_2$,

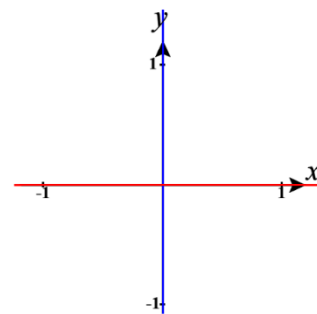
then $\lim_{(x,y) \rightarrow (a,b)} f(x,y)$ does not exist.

Example 1: Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Suppose we approach $(0,0)$ along the x axis. On the x -axis the y -coordinate is always 0. So, the limit becomes $\lim_{(x,0) \rightarrow (0,0)} \frac{x^2 - 0^2}{x^2 + 0^2} = 1$, and $\frac{x^2 - y^2}{x^2 + y^2} \rightarrow 1$ as $(x,y) \rightarrow (0,0)$ along the x axis.

Now, suppose we approach $(0,0)$ along the y axis. On the y -axis the x -coordinate is always 0. So, the limit becomes $\lim_{(0,y) \rightarrow (0,0)} \frac{0^2 - y^2}{0^2 + y^2} = -1$, and $\frac{x^2 - y^2}{x^2 + y^2} \rightarrow -1$ as $(x,y) \rightarrow (0,0)$ along the y axis.

Since $1 \neq -1$, we know that this limit does not exist.



■

Example 2: Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^3}{x^2 + y^6}$ does not exist.

If we approach (0,0) along the y axis, the limit becomes: $\lim_{(0,y) \rightarrow (0,0)} \frac{0 \cdot y^3}{0^2 + y^6} = 0$.

If we approach (0,0) along the x axis the limit becomes: $\lim_{(x,0) \rightarrow (0,0)} \frac{x \cdot 0^3}{x^2 + 0^6} = 0$

(this does not help us because it agrees with the first limit that we found).

Suppose we approach (0,0) along the line $y = mx$ where m is some real number. Then the limit becomes:

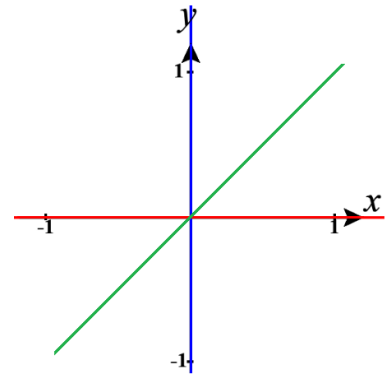
$$\lim_{(x,mx) \rightarrow (0,0)} \frac{x \cdot (mx)^3}{x^2 + (mx)^6} = \lim_{(x,mx) \rightarrow (0,0)} \frac{m^3 x^4}{x^2 + m^6 x^6} \cdot \frac{1/x^2}{1/x^2} = \lim_{(x,mx) \rightarrow (0,0)} \frac{m^3 x^2}{1 + m^6 x^4} = 0$$

(this again does not help us because it agrees with the first limit that we found).

At this point it seems we need to approach (0,0) along a curve which will make the degree of the polynomial on top agree with the degree of the polynomial on bottom. Suppose we approach (0,0) along the curve $x = y^3$,

then the limit becomes $\lim_{(y^3,y) \rightarrow (0,0)} \frac{y^3 \cdot y^3}{(y^3)^2 + y^6} = \lim_{(y^3,y) \rightarrow (0,0)} \frac{y^6}{2y^6} = \frac{1}{2}$.

Since $\frac{xy^3}{x^2 + y^6} \rightarrow 0$ as $(x,y) \rightarrow (0,0)$ along the y -axis, and $\frac{xy^3}{x^2 + y^6} \rightarrow \frac{1}{2}$ as $(x,y) \rightarrow (0,0)$ along $x = y^3$, we know that the limit does not exist. ■



Note: In the above example it was not necessary to calculate the limit along the x axis or along $y = mx$. However, in practice one has to be aware that he or she might check some paths that do not help in proving that a limit does not exist.

Note: In the above example we actually showed that the limit is 0 along infinitely many paths since m could take on infinitely many values. However, the limit STILL DID NOT EXIST because we could find some path that disagreed with the 0 limit. So, in this class checking paths will never lead to a justification that a limit exists.

We now turn to examples where the limit might indeed exist.

Extension of Limit Laws:

Suppose that c is a constant and the limits $\lim_{(x,y) \rightarrow (a,b)} f(x,y)$ and $\lim_{(x,y) \rightarrow (a,b)} g(x,y)$ exist. Then,

1. $\lim_{(x,y) \rightarrow (a,b)} (f(x,y) \pm g(x,y)) = \lim_{(x,y) \rightarrow (a,b)} f(x,y) \pm \lim_{(x,y) \rightarrow (a,b)} g(x,y)$
2. $\lim_{(x,y) \rightarrow (a,b)} [cf(x,y)] = c \lim_{(x,y) \rightarrow (a,b)} f(x,y) \text{ \& } \lim_{(x,y) \rightarrow (a,b)} [c] = c$
3. $\lim_{(x,y) \rightarrow (a,b)} (f(x,y) \cdot g(x,y)) = \lim_{(x,y) \rightarrow (a,b)} f(x,y) \cdot \lim_{(x,y) \rightarrow (a,b)} g(x,y)$
4. $\lim_{(x,y) \rightarrow (a,b)} x = a$
5. $\lim_{(x,y) \rightarrow (a,b)} y = b$
6. $\lim_{(x,y) \rightarrow (a,b)} \sqrt[n]{f(x,y)} = \sqrt[n]{\lim_{(x,y) \rightarrow (a,b)} f(x,y)}$ when n is a positive integer and f is nonnegative near (a,b) when n is even.

Example 3: Find $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - y^4}{x^2 + y^2}$ if it exists.

We note that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - y^4}{x^2 + y^2} = \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 - y^2)(x^2 + y^2)}{x^2 + y^2} = \lim_{(x,y) \rightarrow (0,0)} (x^2 - y^2)$. Now, by the limit laws,

$\lim_{(x,y) \rightarrow (0,0)} (x^2 - y^2) = \lim_{(x,y) \rightarrow (0,0)} x^2 - \lim_{(x,y) \rightarrow (0,0)} y^2 = \left(\lim_{(x,y) \rightarrow (0,0)} x \right)^2 - \left(\lim_{(x,y) \rightarrow (0,0)} y \right)^2 = 0^2 - 0^2 = 0$. Thus, the limit exists and equals 0.

Note: If you tried to prove that the above limit does not exist, you would be unsuccessful because you would get a limit of 0 regardless of the path that you tried. From this point forward you do not need to be as detailed when applying the limit laws. Specifically, you may apply multiple limit laws at the same time.

Note: The instructor may illustrate the above example on the three dimensional plotter if time permits.

Example 4: Find $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 25} - 5}$ if it exists.

First, we note that:

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 25} - 5} &= \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 + y^2)}{(\sqrt{x^2 + y^2 + 25} - 5)} \left[\frac{\sqrt{x^2 + y^2 + 25} + 5}{\sqrt{x^2 + y^2 + 25} + 5} \right] \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 + y^2)(\sqrt{x^2 + y^2 + 25} + 5)}{x^2 + y^2} = \lim_{(x,y) \rightarrow (0,0)} (\sqrt{x^2 + y^2 + 25} + 5). \end{aligned}$$

Finally, by the limit laws: $\lim_{(x,y) \rightarrow (0,0)} (\sqrt{x^2 + y^2 + 25} + 5) = \sqrt{0^2 + 0^2 + 25} + 5 = 10$.

Thus, the limit exists and equals 10.

Now that we are starting to understand the notion of two variable limits, we can introduce the concept of continuity.

Definition: A function f of two variables is called continuous at (a, b) if $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$.

We say f is continuous on D if it is continuous at every point (a, b) in D .

Remark: Intuitively, the meaning of continuity is that if the point (x, y) changes by a small amount, then the value of $f(x, y)$ changes by a small amount. Graphically if $f(x, y)$ is continuous for all (x, y) , the surface determined by $f(x, y)$ has no holes or breaks.

Definition: A polynomial function of two variables is a sum of terms of the form $cx^m y^n$, where c is a constant and m and n are nonnegative integers. A rational function is a ratio of polynomials.

Ex: $f(x, y) = x^3 y - 2yx + 3y^2 - x$ is a polynomial function of two variables.

$g(x, y) = \frac{2xy + 1}{x^2 + y^2}$ is a rational function.

Remark: The extension of the limit laws immediately imply that all polynomial functions are continuous on $\mathbb{R}^2$, and all rational functions are continuous on their domain. This observation makes evaluating some limits very simple.

Example 5: Answer each of the following questions.

a) Find $\lim_{(x,y) \rightarrow (-2,2)} x^3 y - 2yx + 3y^2 - x$ if it exists.

Suppose we let $f(x, y) = x^3 y - 2yx + 3y^2 - x$. Since polynomials are continuous we have that

$$\lim_{(x,y) \rightarrow (-2,2)} f(x, y) = f(-2, 2) = (-2)^3(2) - 2(2)(-2) + 3(2)^2 + 2 = -16 + 8 + 12 + 2 = 6.$$

b) Find $\lim_{(x,y) \rightarrow (1,1)} \frac{2xy + 1}{x^2 + y^2}$ if it exists.

Suppose we let $g(x, y) = \frac{2xy + 1}{x^2 + y^2}$. Since $(1, 1)$ is in the domain of g , we know that g is continuous at

$$(1, 1) \text{ and } \lim_{(x,y) \rightarrow (1,1)} g(x, y) = g(1, 1) = \frac{3}{2}.$$

Question: Could we have used the above strategy to find $\lim_{(x,y) \rightarrow (0,0)} \frac{2xy+1}{x^2+y^2}$?

c) Is $g(x,y) = \frac{2xy+1}{x^2+y^2}$ continuous at $(0,0)$?

No, it is not. In fact $(0,0)$ is not even in the domain of g which means it is not possible for $\lim_{(x,y) \rightarrow (0,0)} g(x,y) = g(0,0)$ to hold.

Example 6:

a) Is the function $f(x,y) = \frac{x^4 - y^4}{x^2 + y^2}$ continuous at $(0,0)$? Explain.

No, it is not continuous at $(0,0)$ since it is not defined at $(0,0)$ (which means that $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = f(0,0)$ does not hold).

b) Is the function $g(x,y) = \begin{cases} \frac{x^4 - y^4}{x^2 + y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$ continuous at $(0,0)$? Explain.

We already showed in an earlier example that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - y^4}{x^2 + y^2} = 0$. Thus, we have that

$\lim_{(x,y) \rightarrow (0,0)} g(x,y) = 0 = g(0,0)$, and $g(x,y)$ is continuous at $(0,0)$ by definition.

Remark: It can be shown (with a careful epsilon delta proof) that if f is a continuous function of two variables and g is a continuous function of a single variable that is defined on the range of f , then the composite function $h = g \circ f$ defined by $h(x,y) = g(f(x,y))$ is also a continuous function.

Note: The above remark can be quite useful in explaining why and where certain two variable functions are continuous.

Example 7: Where is the function $h(x,y) = \arctan\left(\frac{y}{x}\right)$ continuous?

We know from calculus one that the function $g(t) = \arctan(t)$ is continuous for all real numbers. We know that $f(x,y) = \frac{y}{x}$ is a rational function and is therefore continuous everywhere in $\mathbb{R}^2$ except on the line $x = 0$.

We notice that $h(x,y) = g \circ f$. So, by the above remark we know that $h(x,y)$ is continuous everywhere in $\mathbb{R}^2$ except on the line $x = 0$.

As one might expect, the definition of limit of a function of two variables generalizes.

Definition: If f is a function defined on a subset D of $\mathbb{R}^n$ then $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = L$ means that for every $\varepsilon > 0$ there is a corresponding $\delta > 0$ such that if $\mathbf{x} \in D$ and $0 < |\mathbf{x} - \mathbf{a}| < \delta$ then $|f(\mathbf{x}) - L| < \varepsilon$.

Remark: Like the two variable limit we need the limit to agree as we approach $\mathbf{a}$ from all possible directions.

This means that if we want to prove a limit does not exist we must consider curves in $\mathbb{R}^n$. For example, if we wanted to see what the limit approaches along the z -axis, we would consider the limit along the curve $\mathbf{r}(t) = \langle 0, 0, t \rangle$.

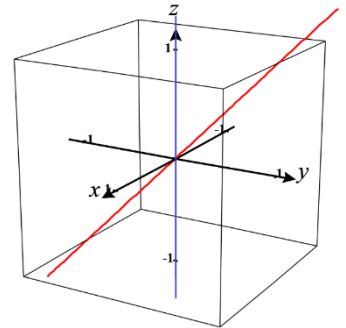
Example 8: Show that the following limit does not exist: $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{yz}{x^2 + 4y^2 + 9z^2}$.

If we approach $(0,0,0)$ along the curve $\mathbf{r}_1(t) = \langle 0, 0, t \rangle$, the limit becomes

$$\lim_{(0,0,t) \rightarrow (0,0,0)} \frac{0 \cdot t}{0^2 + 4 \cdot 0^2 + 9t^2} = 0.$$

If we approach $(0,0,0)$ along the curve $\mathbf{r}_2(t) = \langle 0, t, t \rangle$, the limit becomes

$$\lim_{(0,t,t) \rightarrow (0,0,0)} \frac{t \cdot t}{0^2 + 4t^2 + 9t^2} = \lim_{(0,t,t) \rightarrow (0,0,0)} \frac{t^2}{13t^2} = \frac{1}{13}.$$



Since the limit does not agree along these two curves we know that the limit does not exist.

Note: L'Hopital's Rule does not apply for multivariate functions!

Example 9: Evaluate $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{1 - \cos(x^2 + y^2 + z^2)}{x^2 + y^2 + z^2}$.

If we let $t = x^2 + y^2 + z^2$, then we see that as $(x, y, z) \rightarrow (0, 0, 0)$ it is the case that $t \rightarrow 0^+$.

Therefore, we can say that $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{1 - \cos(x^2 + y^2 + z^2)}{x^2 + y^2 + z^2} = \lim_{t \rightarrow 0^+} \frac{1 - \cos(t)}{t}$

Now since we have reduced the limit to a limit in one variable we see that L'Hopital's Rule applies and we calculate that $\lim_{t \rightarrow 0^+} \frac{1 - \cos(t)}{t} \stackrel{H}{=} \lim_{t \rightarrow 0^+} \sin(t) = 0$.

Therefore, $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{1 - \cos(x^2 + y^2 + z^2)}{x^2 + y^2 + z^2} = 0$.